Syllabus ESM 262 - Spring 2026

A bit about me

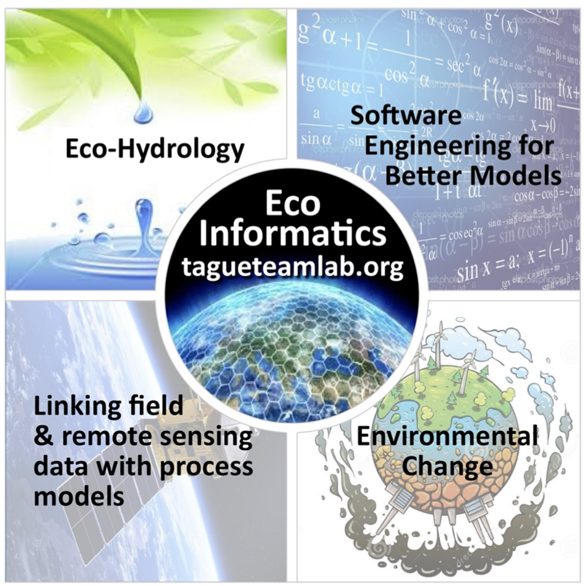

Figure 1: Growing skills

Course description

ESM 262 is an introduction to computing for environmental applications. The course provides practical training in software design best practices.

- programming language concepts
- loops and repetition
- data types and structures

- modular program design;
- reproducibility and version control;
- testing, documentation, and workflow;

The course features **R** for programming, **Git** for version control, **Quarto** for workflow and editing, and **GitHub** for collaboration and publishing, but many concepts would be applicable in other software design tools.

Class will include a mix of lectures and hands-on examples, using students' own computers.

Course where and when

T/TH 8:00 AM - 9:15 AM (Bren Hall 1424)

Teaching Team

Instructor: Naomi Tague (www.tagueteamlab.org)

- **Office hours:** email me
- **email** (tague@ucsb.edu)
- **git user id** naomitague

Teaching assistant: Ojas Sarup

- **Office hours:** Thursday 11am - 1pm (Bren Hall 1005)
- **email** (ojassarup@ucsb.edu)
- **git user id** osarup

Learning objectives

- practice partner coding and reproducible workflows
- 3 *BIG* concepts in programming are *modularity*, *data structure*, *repetition* - we will learn skills related to all three of these
- learn and practice some coding/programming best practices (that will make your data science life easier - whatever you do!)
  - documentation
  - testing
  - reproducibity
- learn some new Rskills that are helpful for a wide variety of data science applications

Computing set up

- Please complete the install-guide on the course website
- Please bring your laptop to class

Course materials and how we will work

All lecture materials will be available on the course website as well as assignments. But you will submit assignments on Canvas

https://naomitague.github.io/ESM_262/

Class time will involve lectures and working on practical applications . This way we use the class time as a lab - where you can get guidance.

Why structure it this way? You learn to code by doing - not so much by watching others do!

Using AI-tools in this course

- AI-tools great resources but effective use requires understanding how to ask good questions
- Use this course to build that understanding, think about the meta concepts.
- Ask yourself - What are situations you might apply a technique/idea/concept;
- You gain skills by **Practicing**, so start without AI-tools
- For assignments/practice
  - **Always** write the code yourself -
  - Supportive ways to use AI-tools
    - use AI-tools to help find syntax
    - Use AI-tools to explain other peoples code (including class examples)
    - Use AI-tools to help interpret error messages

Attendance

- You are expected to come to class - its part of the learning!
- We will regularly have in-class quizzes - These are not really “quizzes” where you answer is right or wrong. They are designed to help you think through the material and provide feedback to me as the instructor. For each quiz you will get full credit for attempting the question in class.
- If you have to miss class let Ojas and me know by email

Tentative Schedule

#	Date	Type	Topic
1	Tue, Apr 14	Regular	Course overview and setup
2	Thu, Apr 16	Regular	Quarto
3	Tue, Apr 21	Regular	Git and GitHub
4	Thu, Apr 23	Regular	Git and Collaboration
5	Tue, Apr 28	Regular	Git a bit more
6	Thu, Apr 30	Regular	Data Types and Structures
7	Tue, May 5	Regular	Looping
8	Thu, May 7	Regular	Modularity
9	Tue, May 12	Evening	Functions!
10	Tue, May 12	Regular	Functions
11	Thu, May 14	Regular	Functions 2
12	Tue, May 19	Evening	Project Design, Modularity and Workflow - Practicing
13	Tue, May 19	Regular	Loops and Functions
14	Thu, May 21	Regular	Modularity-Workflow
15	Tue, May 26	Regular	Testing and debugging
16	Thu, May 28	Regular	Documentation
17	Tue, Jun 2	Regular	Working with APIs/Command Line
18	Thu, Jun 4	Regular	Where does AI fit

Assignments

There will be 6 assignments (more or less one for each week). You will usually have time to start the assignment in class and most will be in groups. We will also have 6 quizes

Assignment expectations and grading

- All assignments submitted on Canvas
- Late Assignments 10% each day. Try to submit on time as we build on the concepts and materials covered in each assignment
- Style counts; So make sure your follow good programming practices including adding documentation and using informative variable names
- You can use AI (ChatGPT etc) to help you learn syntax, but write the code yourself; this is how you learn
- You can resubmit your assignment for additional feedback and we will upgrade your grade. Please do so with in one week of getting the assignment back